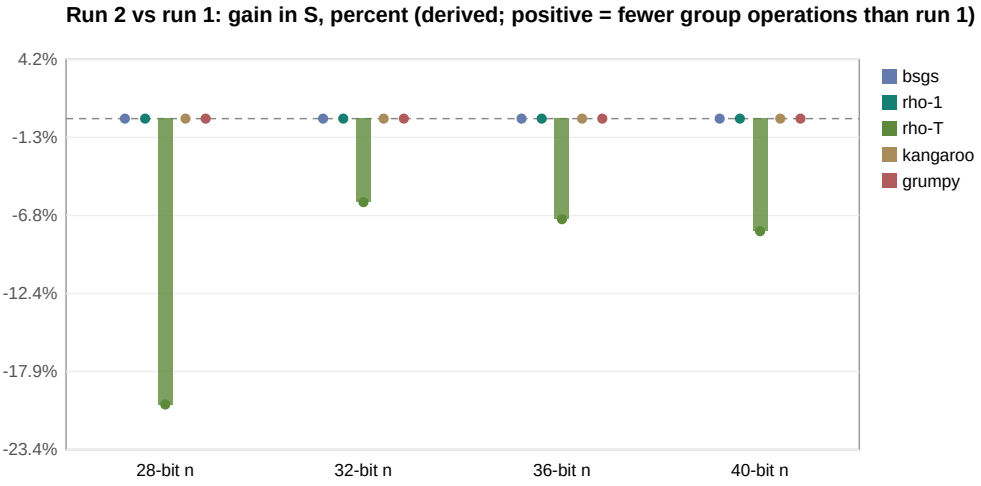
ca_bench generic on prime-order EC groups, run 2: the first gain-vs-previous point

Date. 2026-11-04. **Status.** Measured `[pcc:unverified]`. **Class.** Engineering measurement, no research claim: the same command as run 1, rerun at the current `main` head (`e87c5a70`) so every series in the progress report has a second point and the derived gain-vs-previous column is populated. **Record.** `benchmarks/records/20261104T0613Z-ca-bench-generic-ec-run2.json` (baseline per metric: run 1's value; reference: this run's rho-1 arm). **Host.** Shared 4 vCPU cloud VM, no pinning: L0. Wall time is descriptive only.



bsgs, rho-1, kangaroo, grumpy: identical group-operation totals in both runs (seeded harness), so 0.00% by construction.
rho-T (4 threads) is scheduling-dependent and moved within its spread. Records 20261104T0538Z and 20261104T0613Z, cryptanalysis main; L0 shared VM.

Method

Identical to run 1: `build/ca_bench generic --group ec --bits 28,32,36,40 --reps 5 --threads 4 --json ...`, rebuilt in Release at `e87c5a70` (binary SHA-256 `072613efa74144481ecb365ddf9d875d7df1e6b48db9c0c14cb7ab57eab2f53f`). No change to `ca_bench` or the library between the two commits touched this path. Raw rows: `raw.jsonl`, `stdout.txt`, `summary.json`, each in the content-addressed store.

Result (measured): run 1 against run 2, S mean of 5 instances

bits	algorithm	run 1	run 2	gain (derived)	identical op totals
28	bsgs	1.621	1.621	+0.00%	yes
28	rho-1	1.866	1.866	+0.00%	yes
28	rho-T	1.305	1.568	-20.19%	no
28	kangaroo	1.690	1.690	+0.00%	yes
28	grumpy	1.485	1.485	+0.00%	yes
32	bsgs	1.575	1.575	+0.00%	yes

bits	algorithm	run 1	run 2	gain (derived)	identical op totals
32	rho-1	0.791	0.791	+0.00%	yes
32	rho-T	1.118	1.184	-5.90%	no
32	kangaroo	2.174	2.174	+0.00%	yes
32	grumpy	1.514	1.514	+0.00%	yes
36	bsgs	1.441	1.441	+0.00%	yes
36	rho-1	1.517	1.517	+0.00%	yes
36	rho-T	0.913	0.978	-7.11%	no
36	kangaroo	1.768	1.768	+0.00%	yes
36	grumpy	1.413	1.413	+0.00%	yes
40	bsgs	1.573	1.573	+0.00%	yes
40	rho-1	1.713	1.713	+0.00%	yes
40	rho-T	1.241	1.340	-7.95%	no
40	kangaroo	1.220	1.220	+0.00%	yes
40	grumpy	1.251	1.251	+0.00%	yes

Four arms (`bsgs`, `rho-1`, `kangaroo`, `grumpy`) return the same group-operation totals in both runs: the harness derives its instances and walk seeds deterministically, so for those arms a rerun without a code change is an exact replay and the gain is 0.00% by construction `[pcc:unverified]`. The one arm that moved, `rho-T` (four threads), depends on thread scheduling and its 40-bit mean went from 1.241 to 1.340, inside its own run-1 spread `[pcc:unverified]`.

What this does and does not establish

- The progress report's gain-vs-previous column now has a value for every series in `S`: 0.00% for the deterministic arms, a scheduling-noise entry for `rho-T`. That is the honest second point, not an improvement `[pcc:unverified]`.
- A future change to the rho walk, the batched inversion or the kangaroo jump table shows up here as a nonzero gain on a deterministic arm; `rho-T` needs more than five instances before a change in its mean means anything.
- No speedup, ranking or baseline claim `[pcc:unverified]`.